EvoState: Long-Horizon Evolving State & Inference Scaling

One-Page Scientific Concept & Evaluation Summary | Primary Sources: Kosowski et al. (2025), Engdahl et al. (2026), Gu & Dao (2023), Snell et al. (2024)

DataForge 2026 Pathway Track
Track: Foundation AI & Memory

Status: Verified & Reproducible

CENTRAL FALSIFIABLE SCIENTIFIC CLAIM:

"A fixed-size evolving state can carry useful information across sequences without storing every previous token, but increasing sequence length and conflicting updates can cause interference and information loss; additional inference-time computation can sometimes improve recovery."

1. Theoretical Foundations & Mechanics

- **KV-Cache vs Bounded State:** Autoregressive Transformers maintain O(T·d) KV buffers. Bounded memory maintains O(1) state M_t via outer product binding: M_t = λ M_{t-1} + Δ_t (v_t ⊗ k_t^T).
- **Superposition & Capacity Limits:** Under orthogonal keys, retrieval is exact: v^ = M_T q. When N > α_c·d (Hopfield/Associative limit α_c ≈ 0.14), cross-talk noise causes graceful superposition interference.
- **Inference-Time Scaling (Latent Deliberation):** Test-time refinement optimizes query sharpening without updating model weights: v^{(k+1)} = (1-η)v^{(k)} + η M_T q_{sharp}^{(k)}, helping de-noise superimposed coordinates.

2. Architectural Taxonomy & Comparison

Architecture	State Space	Step Latency	Interference Behavior
Transformer (KV)	O(T·d) Unbounded	O(T) per step	Zero (Exact storage)
Selective SSM (Mamba)	O(d) Bounded	O(1) Constant	Decay over long horizons
BDH (Dragon Hatchling)	O(1) Bounded State	O(1) Constant	Evolving associative state
BDH-CQ (Latent Reason)	O(1) Bounded State	O(K) Variable	Test-time latent deliberation

3. Empirical Sweep Results (2,700 Trials)

- **Delayed Recall (Lag k ∈ [4, 128]):** Educational Toy maintains **100% accuracy** across all lags via selective Δ_t gating, whereas standard vector RNN decays to 0%.
- **Sequence Clutter (T ∈ [32, 512]):** Full History maintains 100% at O(T) latency penalty (7.4ms → 61.1ms). Toy maintains O(1) latency with mild noise accumulation.
- **Destructive Overwrites (N ∈ [1, 8]):** Recency dominance observed; overwrite saturation drops accuracy from 100% (N≤2) to ~45% (N≥5).
- **Subspace Capacity Stress (N_pairs vs d=32):** Abrupt phase transition at N > 4 pairs, matching theoretical associative rank limits.
- **Test-Time Recovery:** Additional inference effort (K=1 to 16) yields measurable recovery when signal is in superposition, but cannot recover fully overwritten state.

4. Evidence Classification Standard

- **[PUBLISHED]** BDH O(N) linear time & state dynamics (Kosowski et al., 2025).
- **[PUBLISHED]** BDH-CQ in-context latent reasoning (Engdahl et al., 2026).
- **[EMPIRICAL]** 2,700 seeded trials in data/precomputed/sweeps_master.csv.
- **[EDUCATIONAL TOY]** d=32 fast-weight surrogate for browser experimentation.

CRITICAL BOUNDARIES, LIMITATIONS & SCIENTIFIC DISTINCTIONS:

1. Educational Toy vs Published BDH: Our browser toy is a pedagogical d=32 associative matrix surrogate. It does NOT implement official multi-billion parameter BDH weights. | **2. Biological Analogy:** Synapse/neuron mathematical duality is illustrative linear algebra; no biological claims are made. | **3. Live vs Precomputed:** Live experiments execute in PyTorch/TypeScript; heavy sweeps are labelled Precomputed with full seed transparency. | **4. Testable Hypothesis:** Tests whether test-time compute improves recovery without asserting universal scaling.

Primary References: [1] Kosowski et al. (2025) *The Dragon Hatchling*, arXiv:2509.26507. [2] Engdahl et al. (2026) *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*, arXiv:2608.09888. [3] Gu & Dao (2023) *Mamba*. [4] Snell et al. (2024) *Scaling LLM Test-Time Compute*. — **Code & Data:** Open Source MIT / Deterministic Seeded (PyTorch 2.0+)